

Autoimmune Connective tissue disease

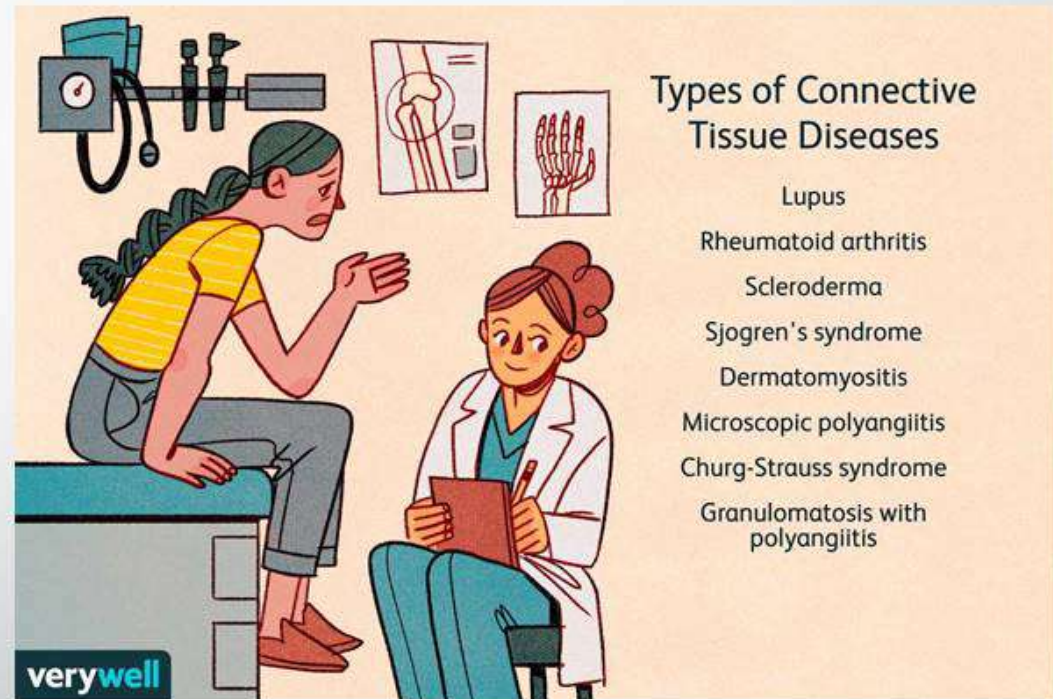
Dr eman gomaa

Consultant dermatology



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- group of autoimmune disorders...
 - Characterized by the production of circulating autoantibodies against one or more of body systems (i.e. self-antigens)._



- **AI-CTDs with cutaneous manifestations:**

1. Lupus erythematosus (LE)

2. Scleroderma

- Diffuse disease (systemic sclerosis [SSc])

- ☆ Limited disease (acrosclerosis, CREST syndrome)

3. Morphea

4. Dermatomyositis (DM)

5. Anti-phospholipid antibody syndrome

6. Mixed connective tissue disease (MCTD)

7. Overlap and undifferentiated CTD

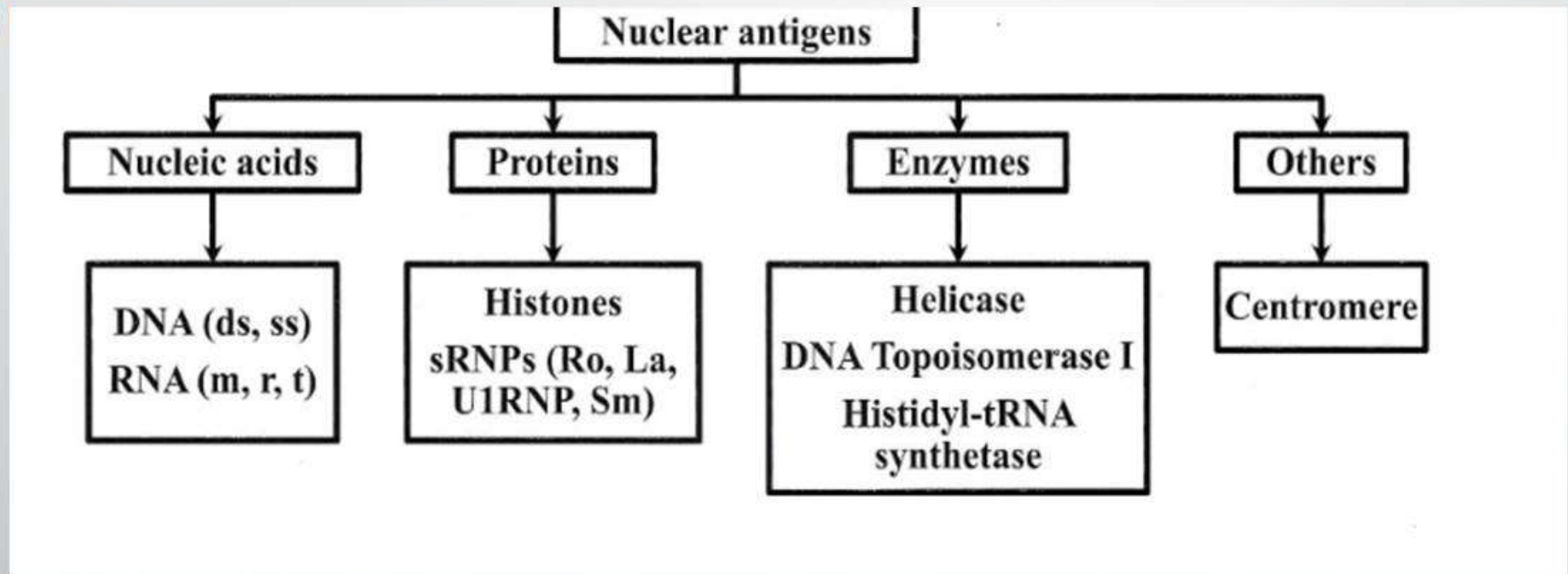
8. Sjogren's syndrome (SS)

9. ANCA-associated vasculitide

- Auto-antibodies in A I-C T D s

These autoantibodies are directed against ALL cellular components

1. Nuclear molecules
2. Cytoplasmic molecules
3. Cell membrane molecules



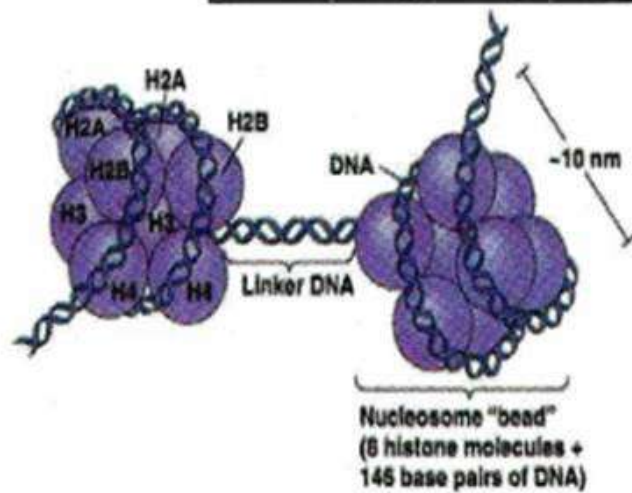


Fig 31.1 DNA double helix wrapped around histone proteins.

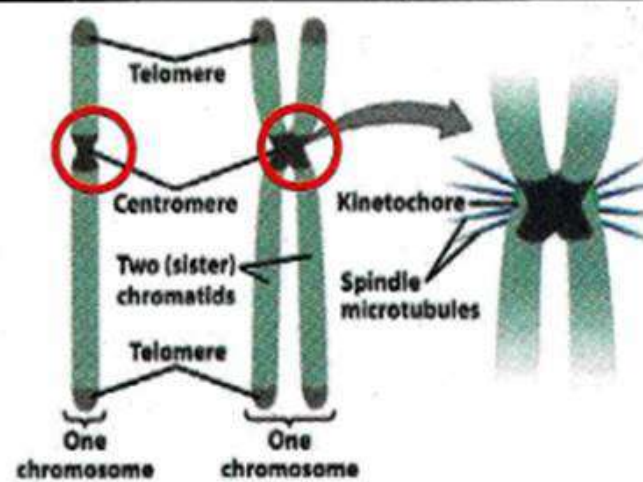


Fig 31.2 Structure of chromosome showing the centromere (red circles).

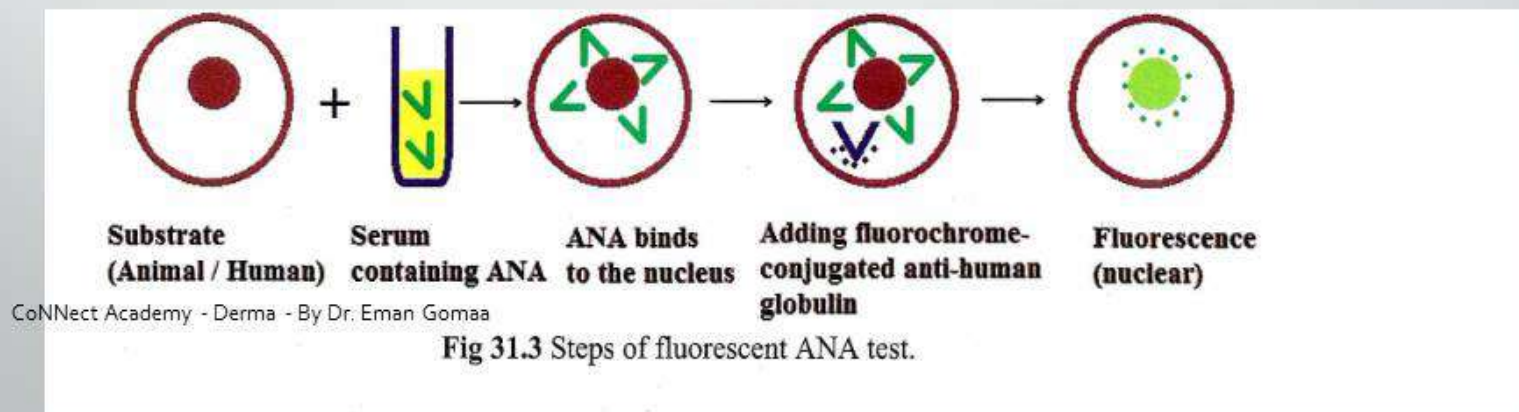
Anti-nuclear antibodies (ANA)

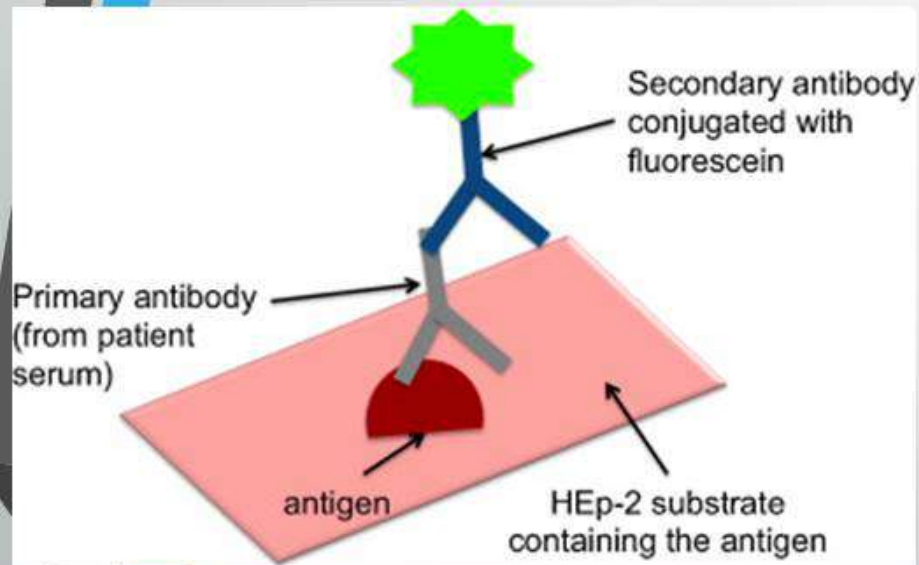


- ANA includes ALL autoantibodies that are reactive against nuclei of human cells.
- ANA are primarily of the IgG class
- ANA are detected by indirect immunofluorescence microscopy , R by enz (ELISA)

FANA: THE CLASSIC ANA ASSAY

- the classic ANA indirect immunofluorescence assay (i.e. FANA) is still the most clinically efficient screening test for systemic autoimmune disorders.
- Identifies antibodies present in serum that bind to autoantigens present in the nuclei of human (mammalian) cells





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- ANA assay substrate:



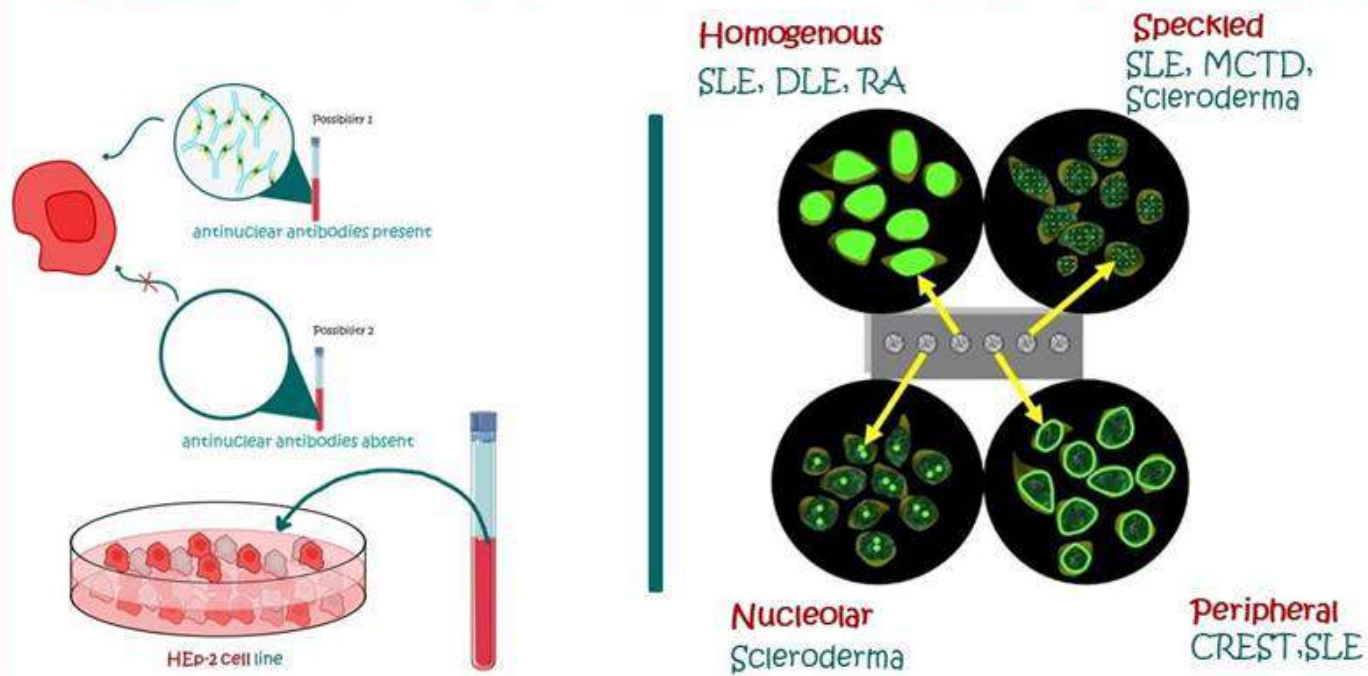
The substrate should be rich in nuclear material. Two types of substrate:

1. Human substrate : human tumor cell line e.g. Hep-2 cells.
2. Animal substrate: rodent cell lines e.g. rat kidney or liver

- Detection:

- Autoantibodies are detected with a fluorochrome-conjugated antiserum (antibodies) that is specific for human immunoglobulin (the autoantibody) that is bound to nuclei in the cell substrate

What does positive ANA means?



ANA assay immunofluorescence patterns:

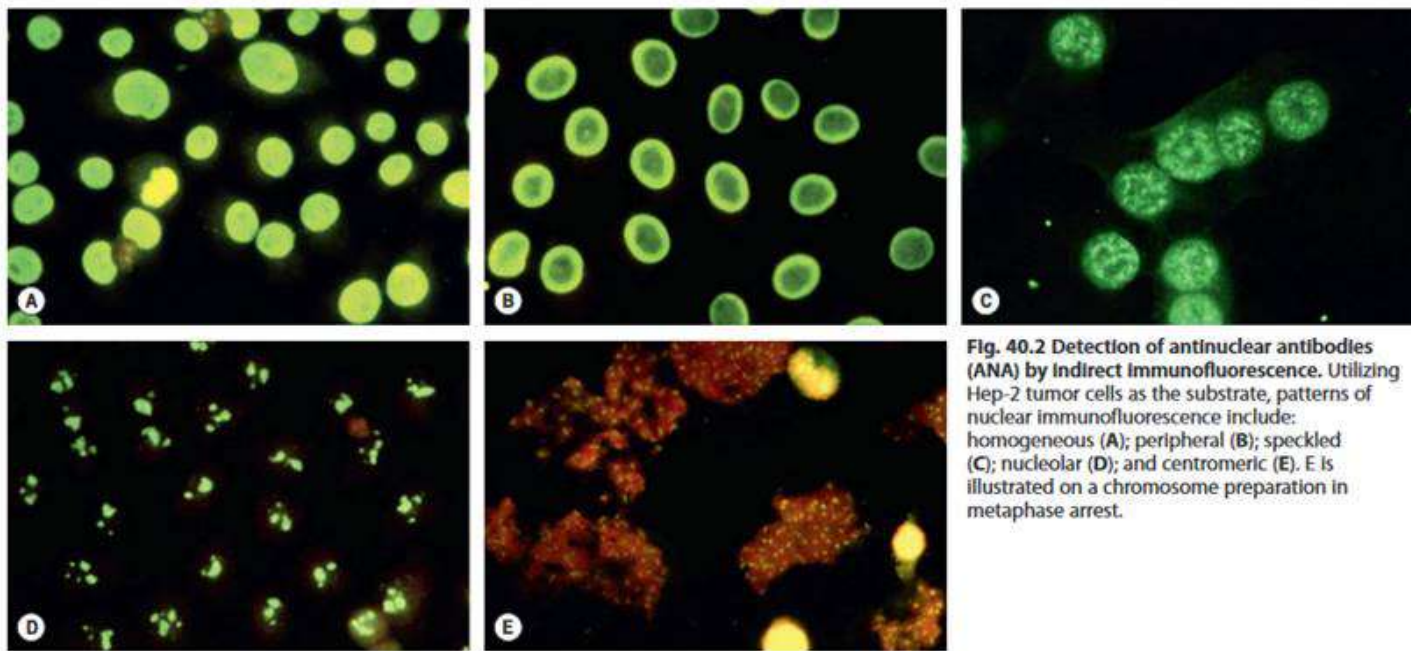


Fig. 40.2 Detection of antinuclear antibodies (ANA) by indirect immunofluorescence. Utilizing Hep-2 tumor cells as the substrate, patterns of nuclear immunofluorescence include: homogeneous (A); peripheral (B); speckled (C); nucleolar (D); and centromeric (E). E is illustrated on a chromosome preparation in metaphase arrest.

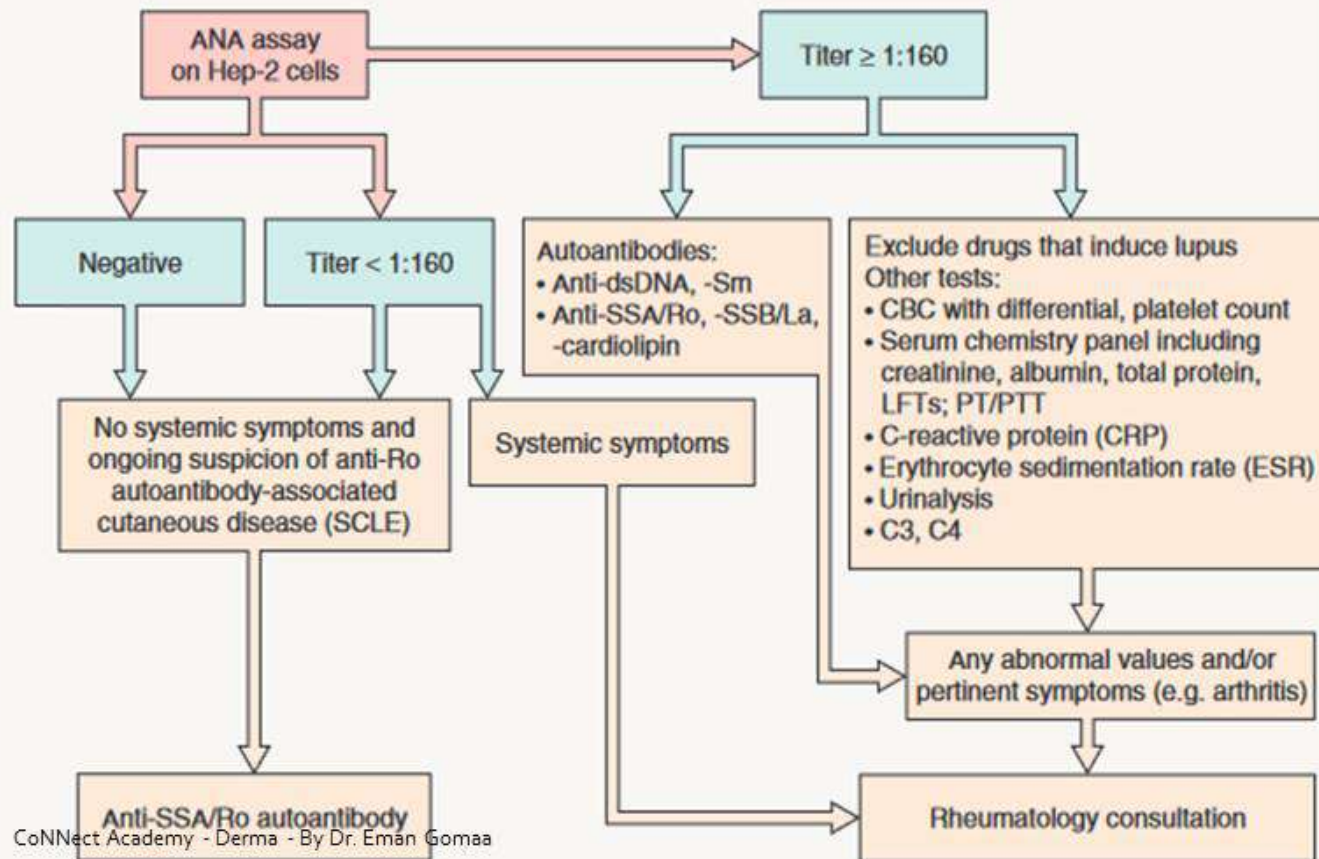
Table 31.1 Patterns of fluorescent ANA test

Pattern	Auto-Abs binding to	Disease
Homogeneous (diffuse)	n(ds)DNA, histones	SLE
Peripheral (rim)	n(ds)DNA	SLE
Nucleolar	Nucleolar RNA	SSc, SLE
Speckled (particulate)	(especially if high-titer) Various sRNPs e.g. Ro/SS-A, La/SS-B, U ₁ RNP, Sm, Scl-70, RNA polymerases (RNAP)	MCTD, SLE, SSc, Sjögren's Syndrome
Centromeric (discrete speckled)	Kinetochore (centromeres)	CREST syndrome = limited form of SSc

- the indications for ordering an ANA test in dermatology

1. Clinical suspicion of AI-CTD (screening)
2. A baseline for patients with discoid lupus erythematosus .
3. Work-up of patients with photosensitivity
4. Baseline for patients undergoing phototherapy
5. Work-up of patients with chronic cutaneous vasculitis

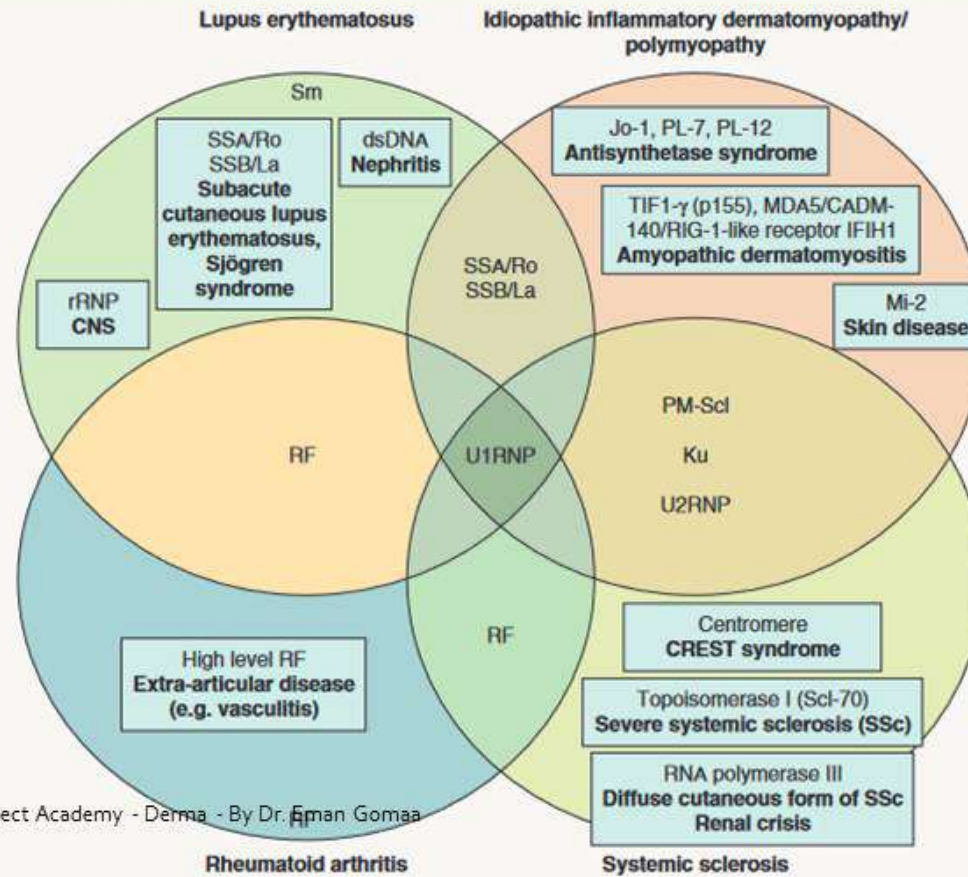
APPROACH TO THE PATIENT WITH SUSPECTED CUTANEOUS LUPUS ERYTHEMATOSUS



Certain autoantibodies have significant disease specificity (diagnostic value)
e.g.

- anti-dsDNA / anti-Sm = highly-specific for SLE
- U1RNP = MCTD
- anti-topoisomerase-1 (Scl-70) = generalized scleroderma
- anti-centromere = CREST syndrome (localized scleroderma)
- anti-Mi-2 = classic dermatomyositis
- anti-Jo-1 = anti-synthetase syndrome

**CORRELATION BETWEEN CLINICAL AND SEROLOGIC MANIFESTATIONS
OF AUTOIMMUNE CONNECTIVE TISSUE DISEASES (AI-CTD)**

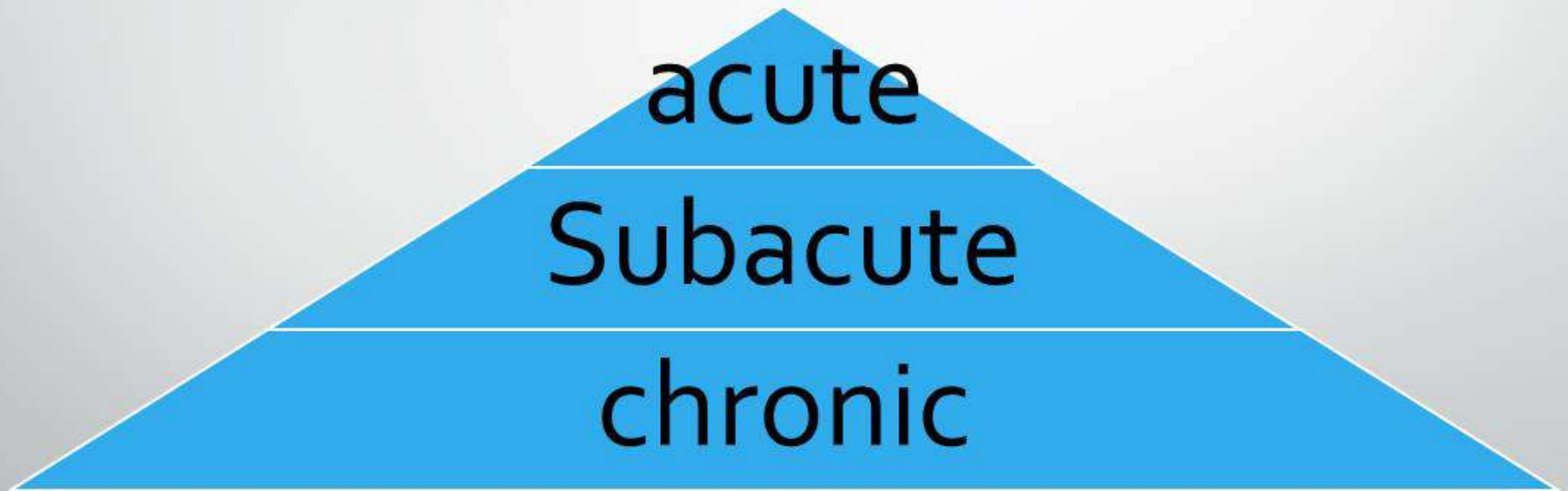


Cutaneous Lupus Erythematosus



- Lupus erythematosus is a multisystem disorder that often affects the skin. Cutaneous lesions , internal disease .

Cutaneous Lupus Erythematosus



Types of cutaneous lupus erythematosus

Acute cutaneous lupus ("acute skin lupus")
"Butterfly rash" (redness across cheeks and nose)



Subacute cutaneous lupus ("subacute lupus")



Red, raised, scaly
nonscarring rash
on sun-exposed
areas

Chronic cutaneous lupus ("discoid lupus")
Red to purple rash with discoloration
and scarring



Scarring and
hair loss

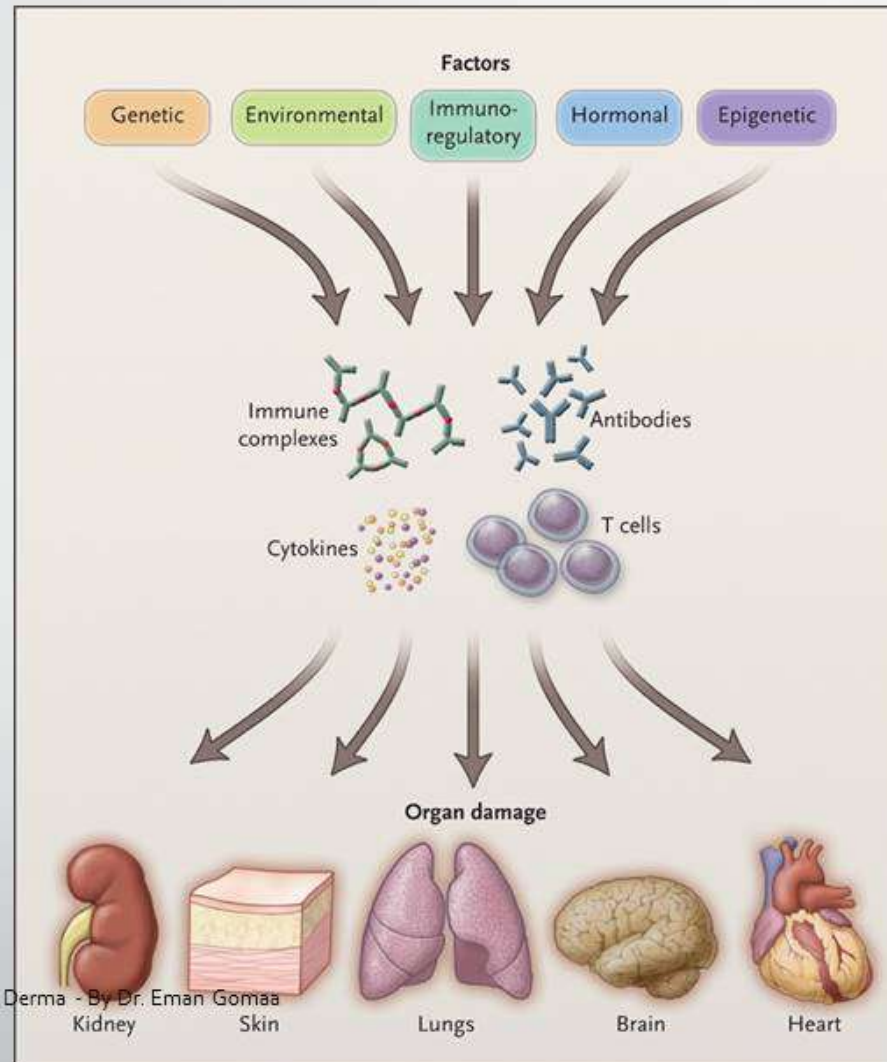
Typical location
(bowl of ear)

Aetiopathogenesis

Enviromental
triggers

In
Geneticly
susptable pt

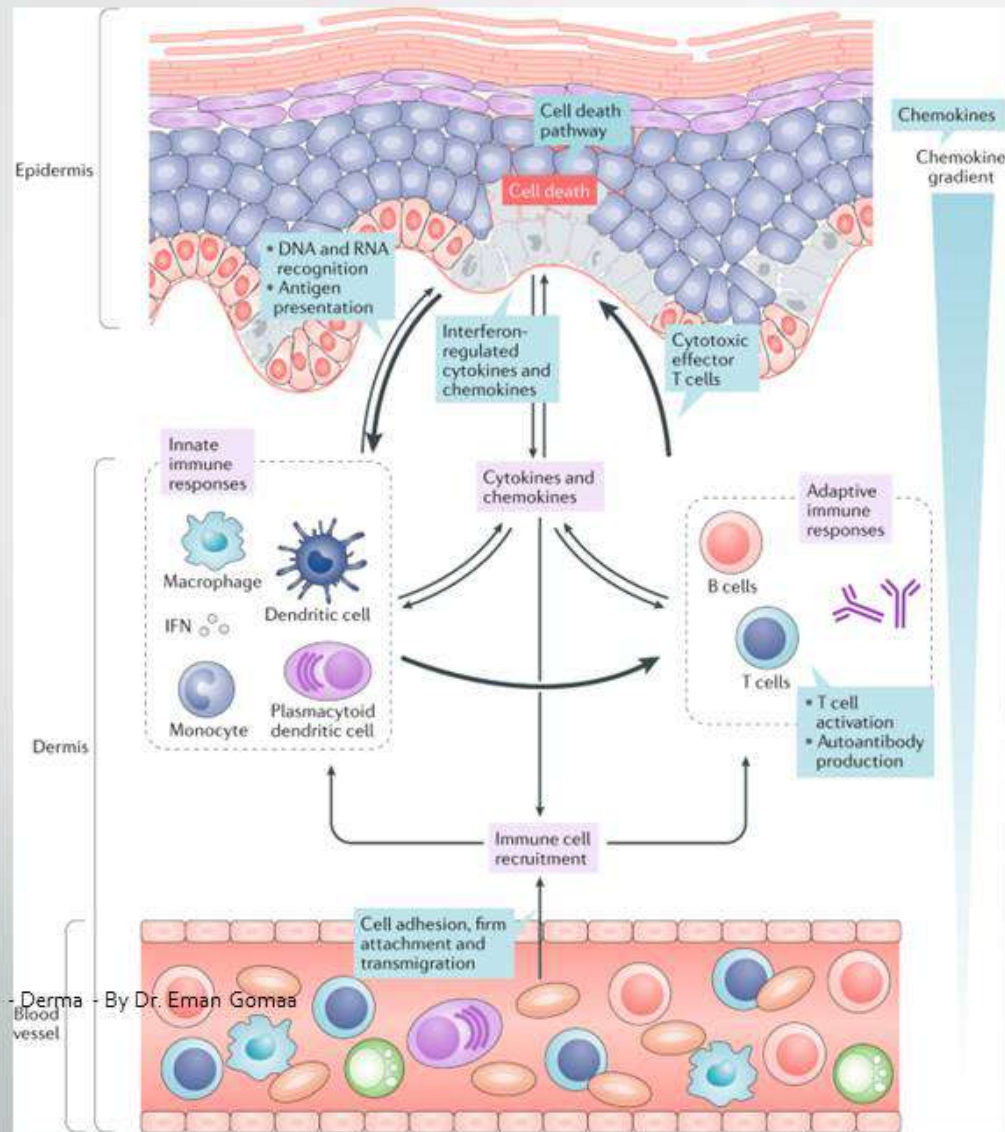
Immune
dysregulation



- **1) Environmental factors:**

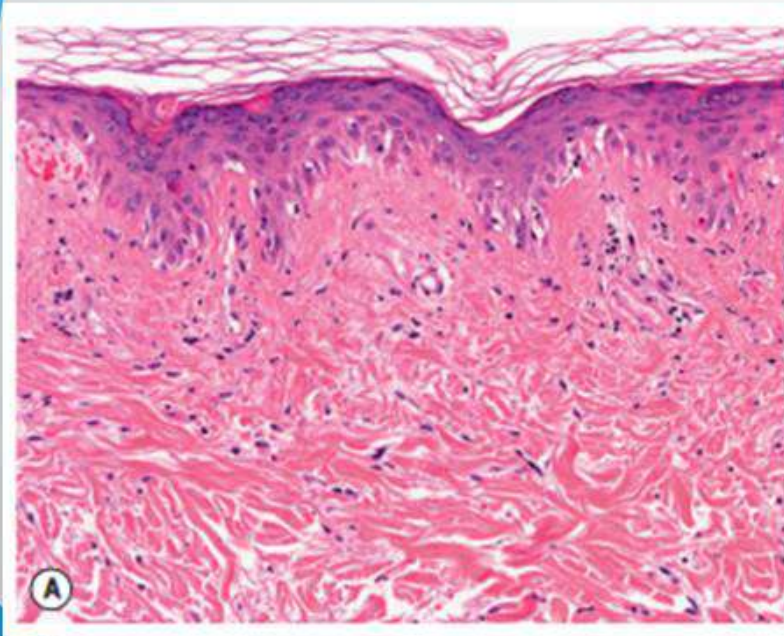
- o **Ultraviolet radiation (UVR):** the strongest trigger .
- o **Medications** (e.g. isoniazid, griseofulvin, penicillamine, dapsone)
- o **Virus infections** (e.g. herpes zoster)
- o **Trauma** (including X rays, diathermy, chemical burns)
- o **Stress**
- o **Seasonal exacerbation** (summer)
- o **Cold exposure**
- o **Pregnancy, Pre-menstrual, Hormone replacement therapy**

- Defect in genes responsible for clearance of apoptotic bodies >>>>immune response >>>>> antigen antibody complex >>>
- evidence:
 - Familial case
 - Association with certain HLA types
 - Role of autoantibodies e.g. anti-Ro / anti-La in SCLE and neonatal lupus



- UVR(mainly UVB) induces apoptosis. Reduction in the clearance of apoptotic cells (due to defects in responsible genes e.g. **STAT4**. signal transducer and activator of transcription 4) ■=>translocation of cellular and nuclear antigens 'A triggers immunity and inflammation

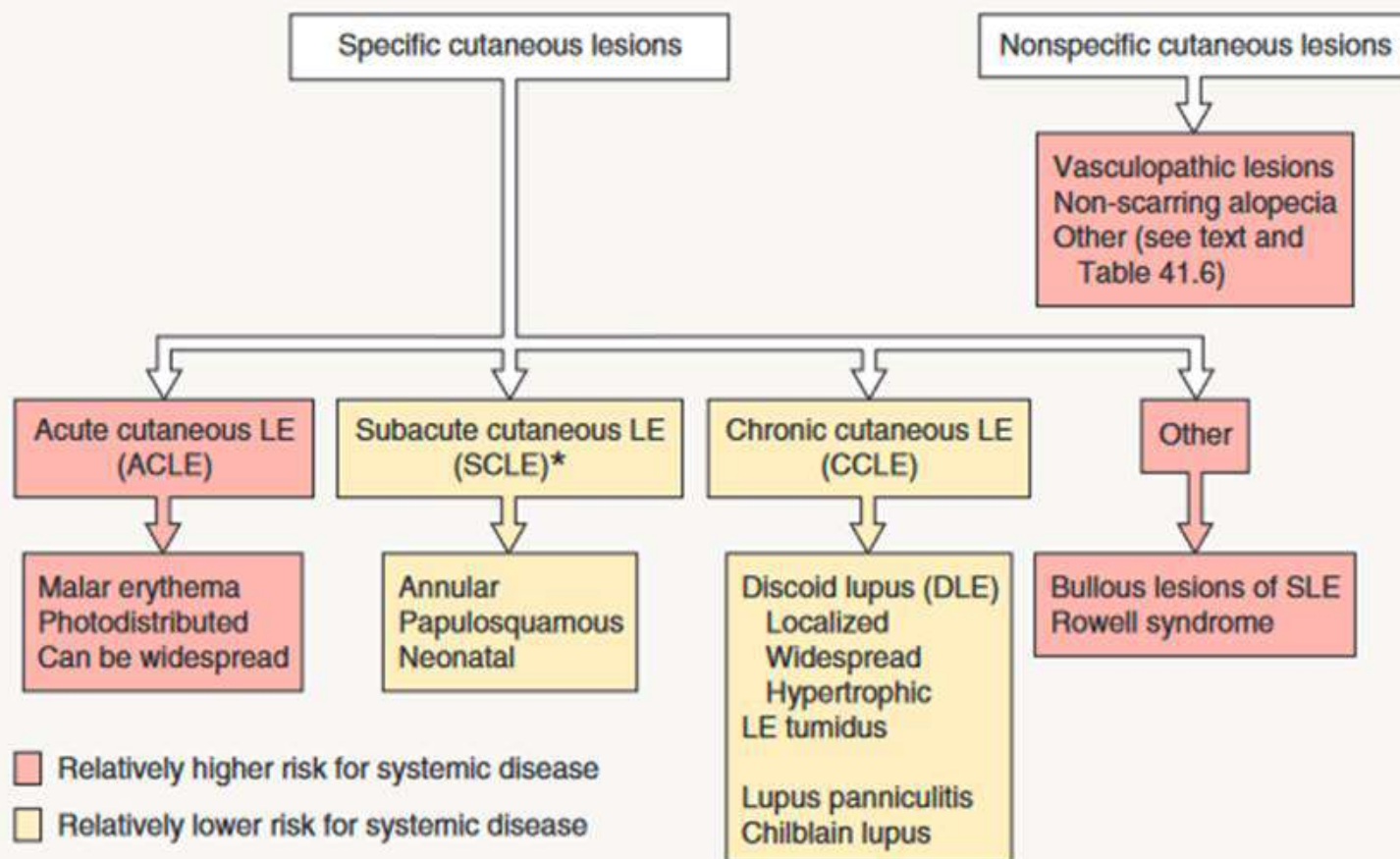
GENES ASSOCIATED WITH SYSTEMIC LUPUS ERYTHEMATOSUS	
B- and T-cell function	<i>HLA-DR</i> – major histocompatibility complex, class II <i>BLK</i> – B lymphoid tyrosine kinase <i>BANK1</i> – B-cell scaffold protein (with ankyrin repeats) 1 <i>CTLA4</i> – cytotoxic T lymphocyte antigen 4 <i>FCGR2B</i> – Fc fragment of IgG, low affinity IIb, receptor (CD32) <i>LYN</i> v-src-1 – Yamaguchi sarcoma viral related oncogene homolog <i>IKZF1</i> – Ikaros <i>PTPN22</i> – protein tyrosine phosphatase, non-receptor type 22 (lymphoid) <i>STAT4</i> – signal transducer and activator of transcription 4 <i>FDCD1</i> – programmed cell death 1 <i>IRAK1</i> – interleukin-1 receptor-associated kinase 1 <i>ETS1</i> – v-ets erythroblastosis virus E26 oncogene homolog 1 <i>TNFSF4</i> – tumor necrosis factor (ligand) superfamily, member 4 <i>IL21</i> – interleukin 21
Innate immunity	<i>IRF5</i> – interferon regulatory factor 5 <i>TNFAIP3</i> – tumor necrosis factor, alpha-induced protein 3 <i>SPP1</i> – secreted phosphoprotein 1 (osteopontin) <i>TREX1</i> – three prime repair exonuclease 1 (see text) <i>TYK2</i> – tyrosine kinase 2 <i>IRAK1</i> – see above <i>STAT4</i> – see above <i>RASGRP3</i> – RAS guanyl releasing protein <i>SLC15A4</i> – solute carrier family 15 member 4
Immune complex clearance	<i>FCGR3A</i> – Fc fragment of IgG, low affinity IIIa, receptor (CD16a) <i>FCGR3B</i> – Fc fragment of IgG, low affinity IIIb, receptor (CD16b) <i>CRP</i> – C-reactive protein, pentraxin-related <i>ITGAM</i> – integrin, alpha M (complement component 3 receptor 3 subunit, CD11b) <i>C4A</i> – complement component 4A <i>C4B</i> – complement component 4B <i>C2</i> – complement component 2 <i>C1Q</i> – complement component 1, q subcomponent
Apoptosis	<i>ATG5</i> – autophagy related 5 homolog <i>STAT4</i> – see above
Ubiquitination	<i>UBE2L3</i> – ubiquitin-conjugating enzyme E2L 3 <i>TNFAIP3</i> – tumor necrosis factor, alpha-induced protein 3 <i>TRAF3</i> – tumor necrosis factor-inducing protein 1
DNA methylation	<i>MECP2</i> – methyl CpG binding protein 2 (Rett syndrome)
Cellular adhesion	<i>ITGAM</i> – see above
Other/unknown	<i>PXK</i> – PX domain containing serine/threonine kinase <i>ICA1</i> – islet cell autoantigen 1



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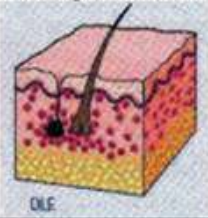
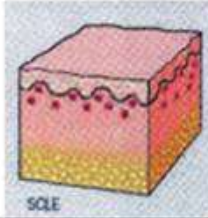
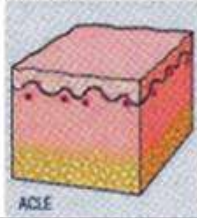
- (3) Inflammation:
- Lichenoid tissue reaction (epidermal basal cell damage and a band-like lymphocytic infiltrate in the upper dermis)

CLASSIFICATION OF CUTANEOUS LUPUS ERYTHEMATOSUS



*In adult patients, drug-induced SCLE should be considered.

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	<i>Chronic cutaneous LE (CCLE)</i>	<i>Subacute cutaneous LE (SCLE)</i>	<i>Acute cutaneous LE (ACLE)</i>
Classic phenotype	Discoid LE (DLE)	A photosensitive eruption	Malar erythema
Other clinical types	* Tumid LE * Lupus panniculitis * Lupus profundus * Chilblain lupus	* Neonatal lupus erythematosus	* Bullous eruption of SLE
	* Lupus / lichen planus overlap syndrome * Rowell's syndrome		
Course	Chronic (long-lasting)	Less chronic (more long-lasting > ACLE)	Transient
Female: male	2 : 1	3 : 1	9 : 1
Age	4 th decade in females, slightly later in males	Adults	Child-bearing period (30's)
Inflammation (intensity and depth)	Intense (superficial & deep dermis, peri-adnexal, perivascular) 	Moderate Superficial dermis 	Minimal Superficial dermis 
Scarring	Permanent disfiguring scars	Minimal or absent	Absent
Association with SLE	5 % MCQ	10-15 (up to 50) %	Almost always

changing characteristics across this spectrum.

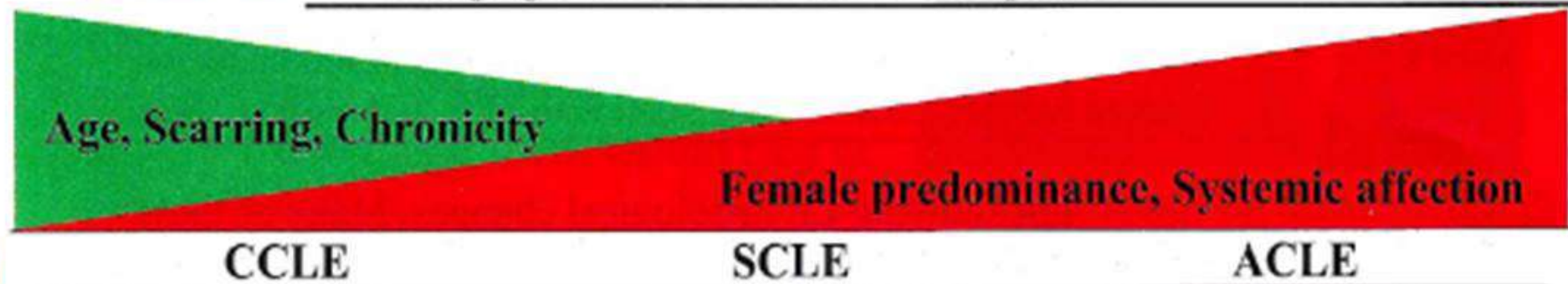


Fig 32.2 Spectrum of *cutaneous* LE.

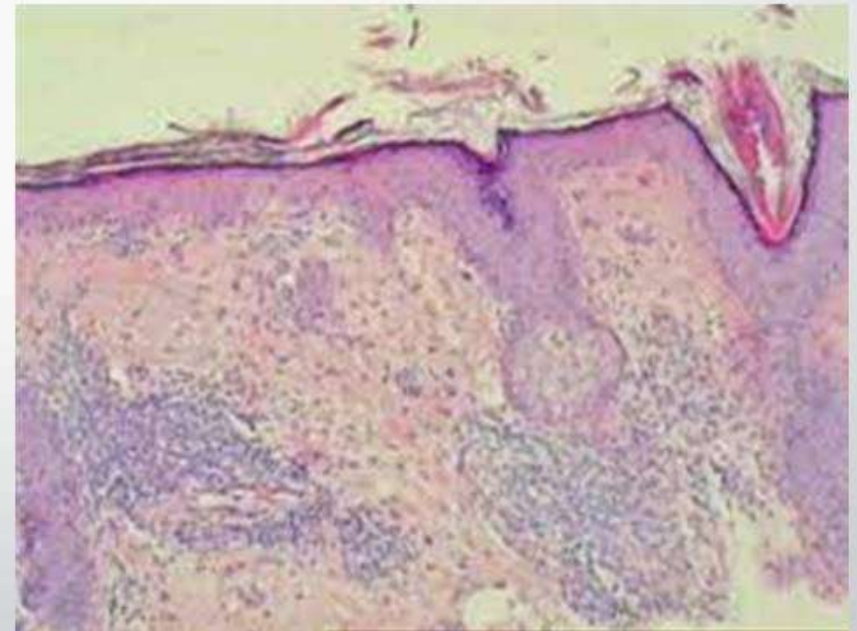
- Vacular interface dermatitis:

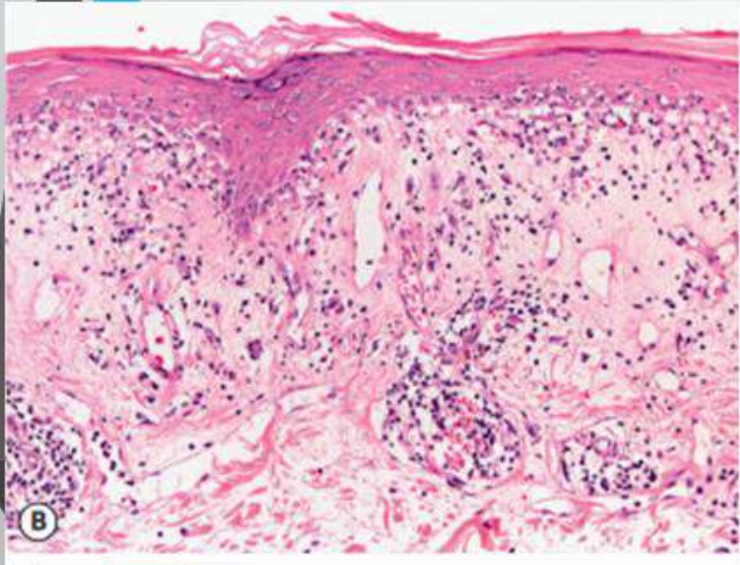
Vacuolar alteration of DEJ with keratinocytes necrosis and accumulation of melanophages in the upper dermis.

Peri-vascular lymphocytic infiltrate (superficial / superficial and deep).

Histopathology of CCLE

1. Hyperkeratosis
2. Follicular plugging
3. Epidermal atrophy
4. Dyskeratosis
5. Hydropic degeneration of DEJ
6. Thicking BM
7. Dermis: patchy lymphocytic infiltrate
Mucin deposition , odema , V.D pigmentary incontinance



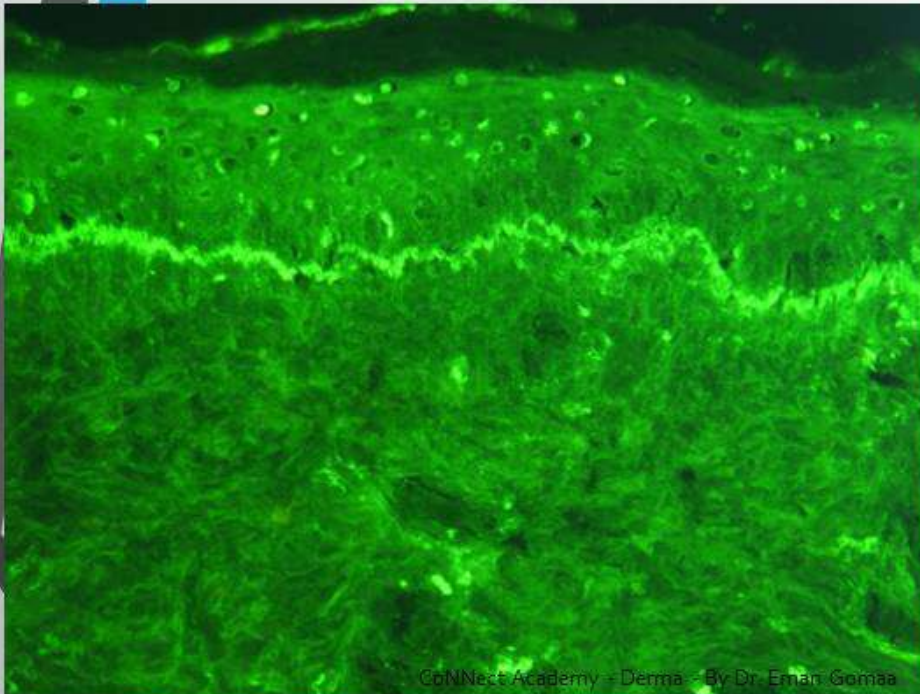


-) Subacute cutaneous LE

- Little or no hyperkeratosis, follicular plugging, basement membrane thickening, periadnexal infiltrate, deep dermal infiltrate, or scarring

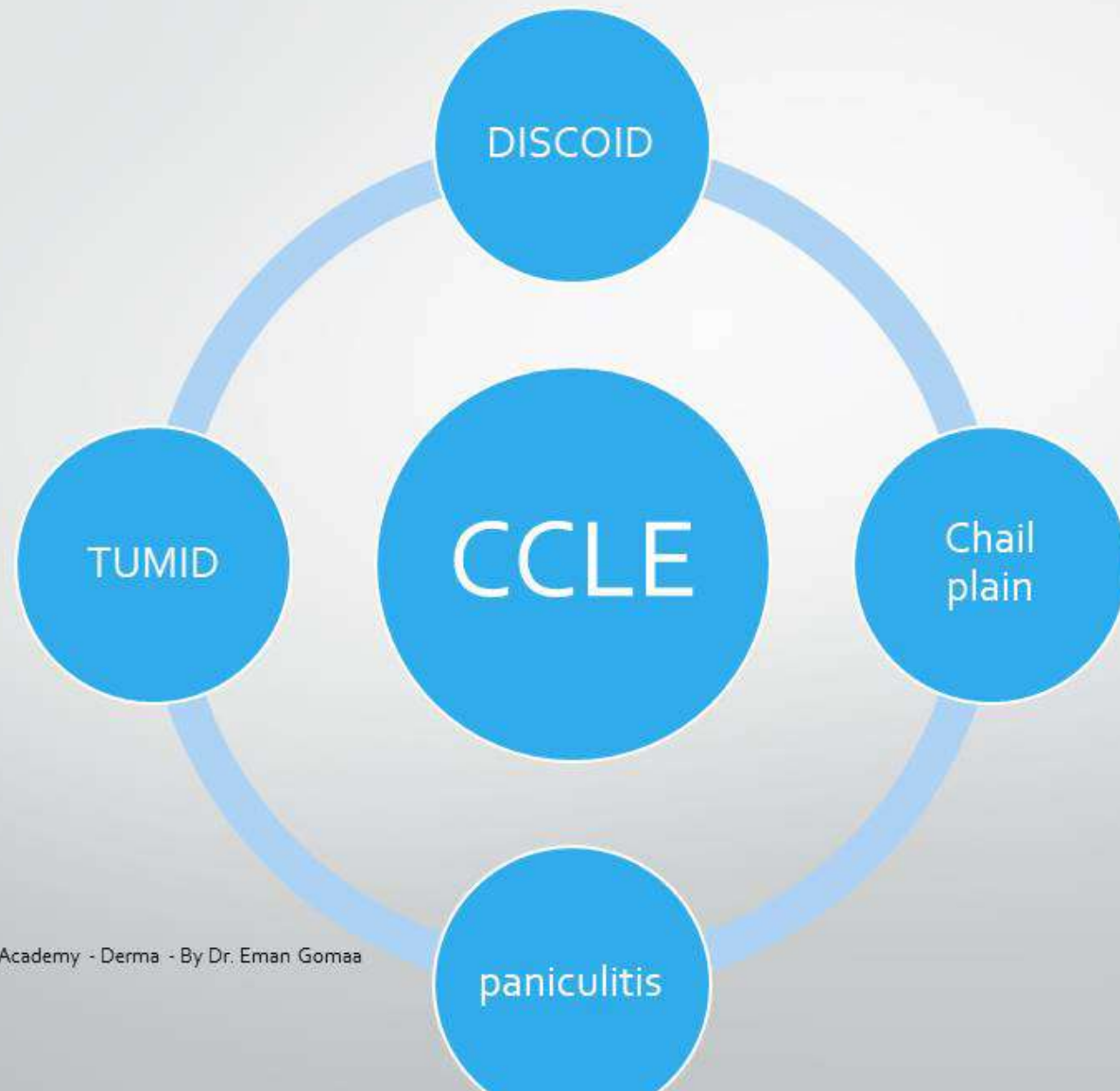
- More intense inflammation at the dermal-epidermal interface, more severe hydropic degeneration (up to formation of clefts and subepidermal vesicles) and more pronounced dermal edema

Lupus band test



DIF make only in pt with equivocal HP

Granular deposition of igG +complement at
DEJ (lupus band)



Discoid LE



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- Discoid erythematous telangiectatic plaque
- Follicular plugging (stipling)
- Scarring , atrophy
- Central hypopigmentation
- Peripheral hypopigmentation
- Site :
 - Localized : head , neck , face ,
 - Disseminated : any area sun exposed , sun protected



- Complication :

- Scar
- Scc
- SLE : 5% OF DLE have SLE , 15% of SLE have DLE

DD



- Sarcoidosis
- Granuloma faciale
- Lupus vulgaris
- Jessner's lymphocytic infiltrate
- Lymphocytoma cutis
- Lymphoma cutis
- Polymorphous light eruption
- Dimorphic fungal infections



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- **A) DD of patchy dermal lymphocytic infiltrate:** (called the five L's).
 1. LE
 2. Lymphocytic lymphoma (atypical lymphocytes in an interstitial distribution “Indian filing”)
 3. Plaque type of polymorphous light eruption (prominent papillary dermal edema)
 4. Lymphocytoma cutis (heavier infiltrate with no tendency to be arranged around pilosebaceous structures)
 5. Lymphocytic infiltration of Jessner

Tumid le

- Erythematous edematous plaque
- Induration, urticaria, erythema
- No scale, no plugging, no scar, no atrophy
- HP :no epidermal
- Intense dermal infiltrate
- DD :urticarial vasculitis
 - 5L

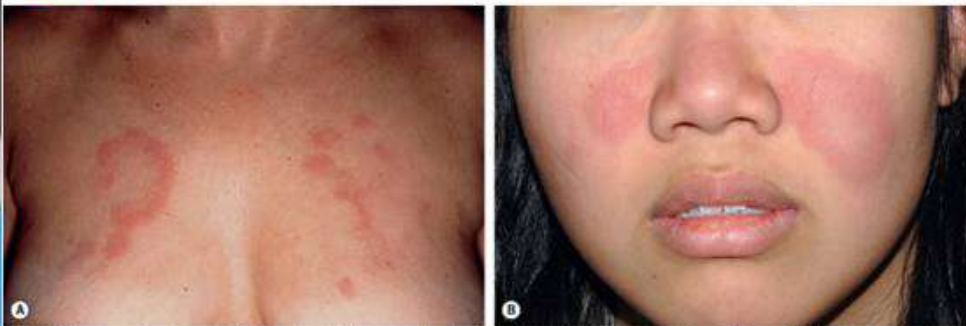


Fig. 41.11 Lupus erythematosus tumidus. Annular pink plaques on the chest (A) and pink-violet plaques on the face (B). None of the lesions have epidermal change. (Courtesy, Julie V. Schaffly, MD)

Lupus panniculitis



Fig. 41.12 Lupus panniculitis. Erythematous plaque on the upper arm. The lesions may resolve with lipoatrophy.



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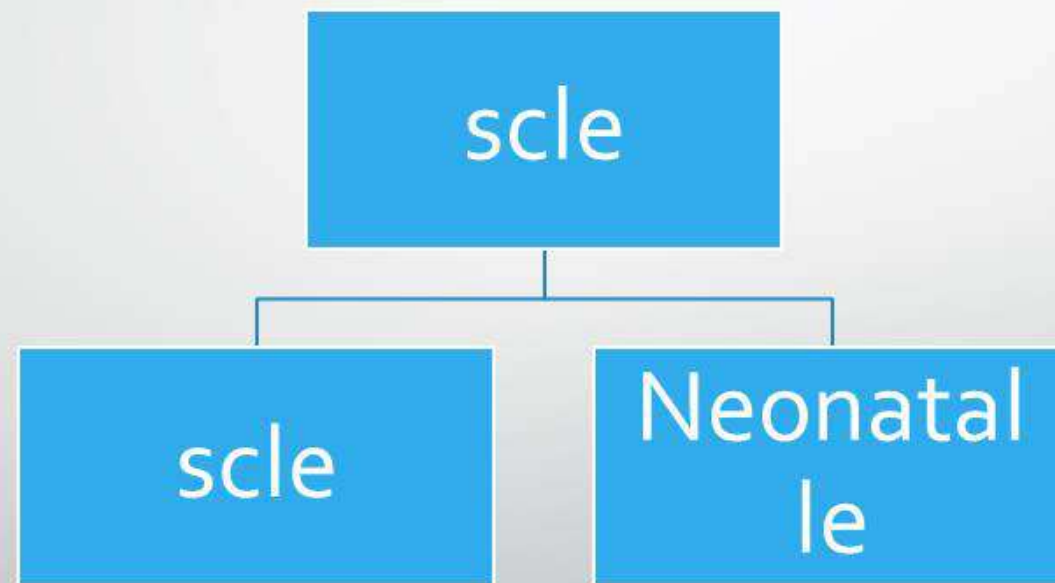
- Intense inflammation of sc.c fat
- Face , upper arm , upper trunk , breast , buttocks
- Firm nodule or plaque under normal skin
- Cup shape
- Discoid lesion over panniculitis >>>>lupus profundus

Chilblain le



Fig. 41.13 Chilblain lupus. Violaceous plaques, some with scale, on toes. If there is a family history of this disorder, the possibility of mutations in *TREX1*, which encodes a DNA exonuclease, or *SAMHD1*, which encodes a host restriction nuclease that plays a role in the innate immune response, can be considered.

- Red purple papule , plaque on toe , finger
- Nose , ear , knee , elbow
- Increase by cold , pregnancy



SCLE



Fig. 41.8 Subacute cutaneous lupus erythematosus (SCLE). Lesions are most commonly seen on the upper trunk and sun-exposed aspects of the upper extremities. The margins of the annular lesions may have scale-crust (A) or be composed of multiple papules (B). C Note the peripheral scale and relative sparing of the proximal interphalangeal joints. C, Courtesy, Lorenzo Geronzi, MD.

- Aetiopathogenesis:
 - Drug: hydrochlorothiazide , terbinafen
- 1. Photosensitivity : sun exposed
- 2. Annular , polycyclic
- 3. Psoriasiform lesion
- No scar , heal with hypo or hyperpigmentation

- 
- DD annular erythematous scaly lesion :
 - Annular ps , tinea circinate , pr , pemphigus foliaceus , GA , annular eczema
 - Serology : anti RO , anti La , anticardiolipin

Neonatal LE

- Infant whose mothers have anti RO –AB
- 3 system
 1. Skin : racoon eyes , photosensitivity , resolve without scar
 2. Congital heart block : 20% mortality
 3. Hepatopiliary , thrombocytopenia
- Investigation : ECO , CBC , liver function



Fig. 41.14 Neonatal lupus erythematosus. Annular erythematous plaques on the forehead and scalp. Note the resemblance to the annular form of subacute cutaneous lupus erythematosus. *Courtesy, Julie V Schaffer, MD.*

- Treatment :
 - cutaneuse : mild no TT
 - Cardiac :pace maker
 - Hepatic , hematological : reversable

Acute SLE

Multiorgan disease

Specific
Malar erythema
Poikiloderma
bullae on top of rash
Mucosal ulcer
Leg ulcer
Eruptive DF
mucinosi

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Non specific
Vascular

1. Raynauds
2. Livedoreticularis
3. Periangual telangiectasia
4. Purpura
 - Scleroderma
 - Calcinosis cutis
 - Rheumatoid nodules
 - Alopecia
- Palmoplantar erythema

- **Systemic manifestations:**

- fever and constitutional symptoms, almost every body system may be affected in SLE e.g. joints (arthralgia and arthritis), musculoskeletal (myalgia), respiratory, gastrointestinal (vasculitis and abdominal pain), renal, hematological (anemia, leucopenia, thrombocytopenia or pancytopenia), and many others.

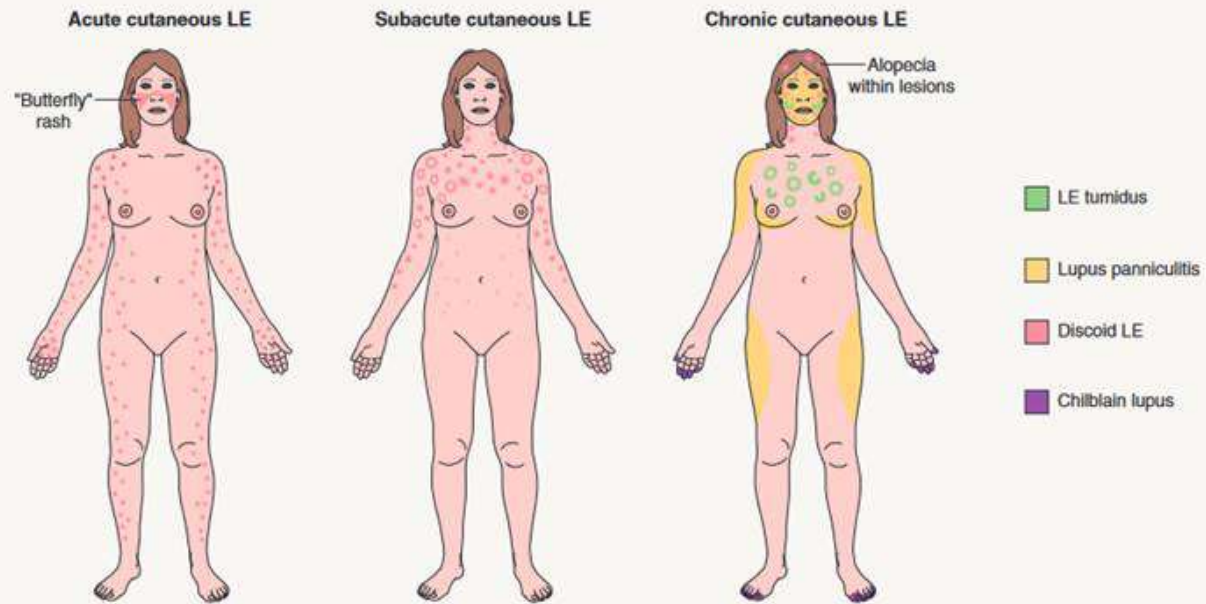


Fig. 41.10 Acute cutaneous lupus erythematosus (ACLE). This patient had ACLE lesions on the arms as well as the face.





CHARACTERISTIC SITES OF INVOLVEMENT FOR THE THREE MAJOR SUBTYPES OF CUTANEOUS LUPUS ERYTHEMATOSUS



1997 UPDATE OF THE 1982 AMERICAN COLLEGE OF RHEUMATOLOGY REVISED CRITERIA FOR CLASSIFICATION OF SYSTEMIC LUPUS ERYTHEMATOSUS*

Criterion	Definition
1. Malar rash	Fixed erythema, flat or raised, over the malar eminences, tending to spare the nasolabial folds
2. Discoid rash	Erythematous raised patches with adherent keratotic scaling and follicular plugging; atrophic scarring may occur in older lesions
3. Photosensitivity	Skin rash as a result of unusual reaction to sunlight, by patient history or physician observation
4. Oral ulcers	Oral or nasopharyngeal ulceration, usually painless, observed by physician
5. Arthritis	Non-erosive arthritis involving two or more peripheral joints, characterized by tenderness, swelling or effusion
6. Serositis	a) Pleuritis – convincing history of pleuritic pain, rubbing heard by a physician, or evidence of pleural effusion OR b) Pericarditis – documented by ECG, rub or evidence of pericardial effusion
7. Renal disorder	a) Persistent proteinuria greater than 0.5 g/day or greater than 3+ if quantitation not performed OR b) Cellular casts – may be red cell, hemoglobin, granular, tubular or mixed
8. Neurologic disorder	a) Seizures – in the absence of offending drugs or known metabolic derangements, e.g. uremia, ketoacidosis or electrolyte imbalance OR b) Psychosis – in the absence of offending drugs or known metabolic derangements, e.g. uremia, ketoacidosis or electrolyte imbalance
9. Hematologic disorder	a) Hemolytic anemia with reticulocytosis OR b) Leukopenia – less than 4000/mm ³ total WBC on two or more occasions OR c) Lymphopenia – less than 1500/mm ³ on two or more occasions OR d) Thrombocytopenia – less than 100000/mm ³ in the absence of offending drugs
10. Immunologic disorder	a) Anti-DNA antibody to native DNA in abnormal titer OR b) Anti-Sm: presence of antibody to Sm nuclear antigen OR c) Positive finding of antiphospholipid antibodies based on: (1) an abnormal serum level of IgG or IgM anti-cardiolipin antibodies; (2) a positive test result for lupus anticoagulant using standard methods; or (3) a false-positive serologic test for syphilis known to be positive for at least 6 months and confirmed by <i>Treponema pallidum</i> immobilization or fluorescent treponemal antibody absorption test (FTA-ABS)

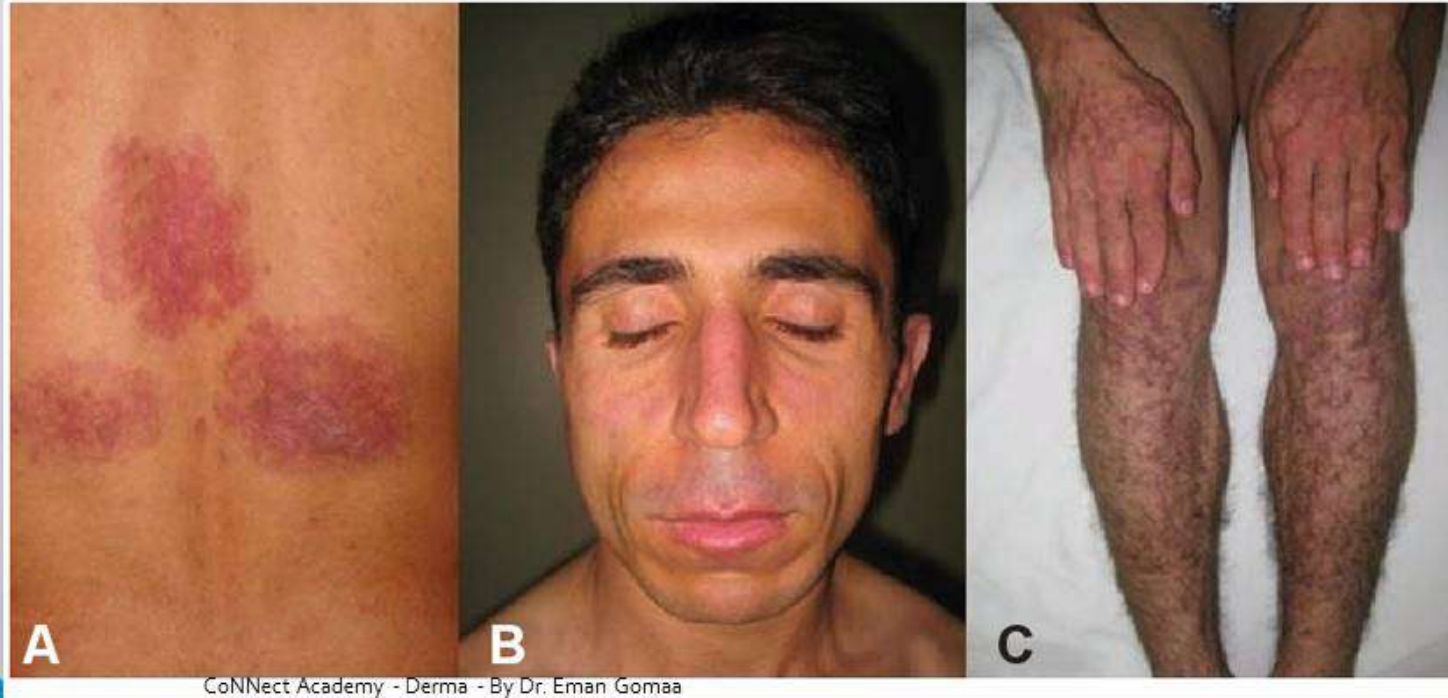
Bullous sle



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- AB against collagen 7
- Hp like DH with neutrophils

Lupus –lp

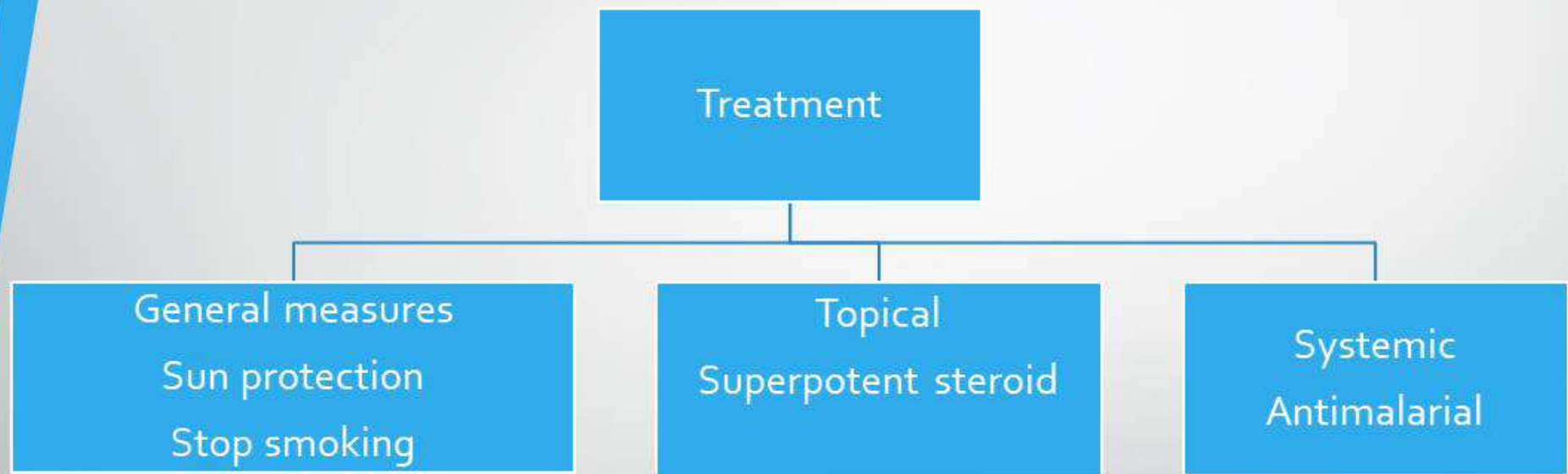


Rowells syndrom



- Cutaneous LE
- Erythema multiform like lesion no face , neck , hand , chest





a) Anti-malarial therapy: (the gold standard for systemic therapy)

	Hydroxychloroquine	Chloroquine	Quinacrine/Mepacrine
Safety	Most safe	More toxic	
Dose	200 – 400 mg/day	125–250 mg/day	
Eye toxicity (maximum daily dose)	Less toxic (6.5 mg/kg ideal body weight)	More toxic (3.5–4 mg/kg ideal body weight)	Quinacrine does <u>not</u> cause eye toxicity (but turns the skin yellow)
Notes	The most commonly used antimalarial		In patients unresponsive to hydroxychloroquine, quinacrine may be added to the regimen

- **treatment of systemic LE**



- NO major organ involvement :

- a) mild disease: NSAIDs

- b) moderate to severe disease: corticosteroids and immunosuppressives .



- Mild degree of major organ involvement: corticosteroids



- Moderate to severe involvement: pulse cyclophosphamide $\pm$ pulse corticosteroids are often recommended



- Immune response modifiers (biologic therapies) are reserved for refractory disease

THERAPY OF CUTANEOUS LUPUS ERYTHEMATOSUS

Rx

Local therapy

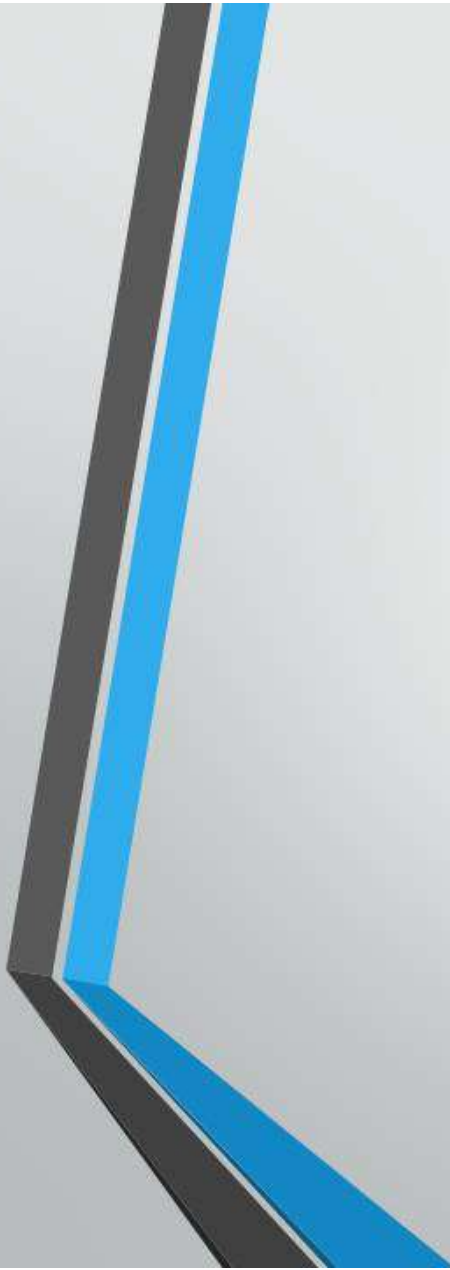
Sun protection (2)
Topical and intralesional corticosteroids (2)
Topical calcineurin inhibitors (2)
Topical retinoids (3)

Systemic antimalarial therapy

Hydroxychloroquine (200 mg po daily–BID in adults; up to 6.5 mg/kg ideal body weight/day) (2)[§]
Chloroquine (125–250 po daily in adults; up to 3.5–4 mg/kg ideal body weight/day) (2)[§]
Quinacrine (100 mg po daily) (2)
Combination of hydroxychloroquine or chloroquine and quinacrine (2)

Systemic therapy for antimalarial-resistant cutaneous disease

Retinoids (e.g. acitretin, isotretinoin) (2)
Thalidomide (50–100 mg po daily for clearing and, if necessary, 25–50 mg po daily–twice weekly for maintenance)^{*} (2)
Lenalidomide (3)
Dapsone (primarily for bullous eruption of SLE) (2)
Immunosuppressive agents (e.g. mycophenolate mofetil, azathioprine, methotrexate) (2)
Sulfasalazine (2)
Systemic corticosteroids (3)



Thank You

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